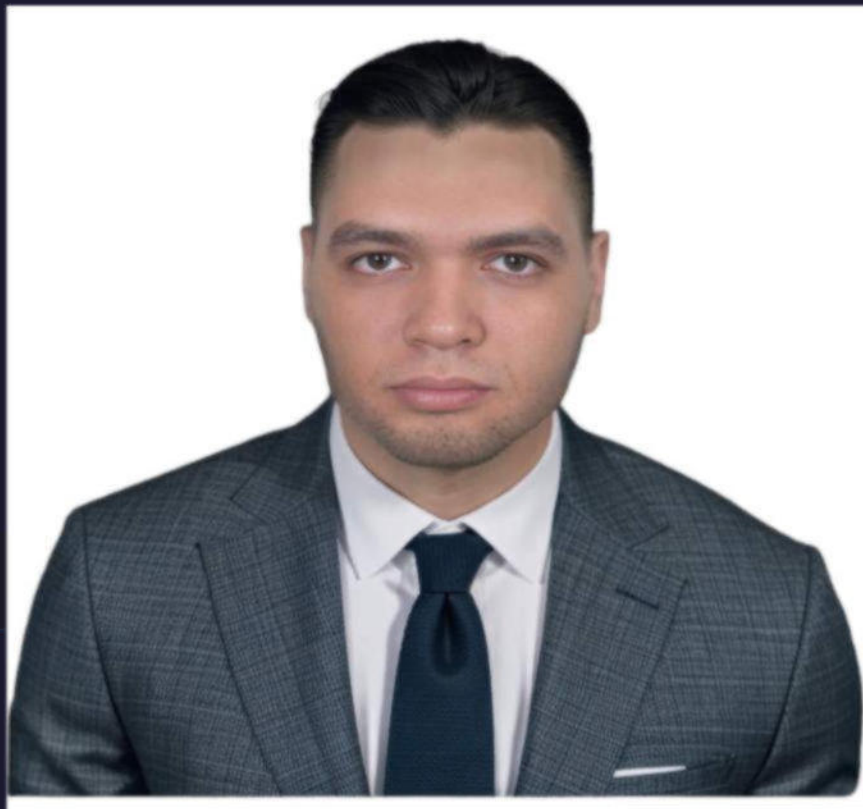


portfolio



✓ Data Analysis

✓ Mohamed Mahmoud Fouad



E-Commerce Customers Personality

Exploratory Data Analysis using SQL to Extract Insights



#EXTRACTED RESULTS :

Y Extract Purchases Details :

- Total Purchases
- Deal Purchases & it's Percentage
- Web Purchases & it's Percentage
- Catalog Purchases & it's Percentage
- Store Purchases & it's Percentage

Y Extract Recency Details :

- Recency Days (Last Purchase)
- Total Customer In Prev 14 Days
- Total Customer 15 To 30 Prev Days
- Total Customer 31 To 60 Prev Days
- Total Customer 61 To 99 Prev Days

Y Calculate Metrics as :

- Top 5 Customer in Income
- Calculate Total Website Visits
- Calculate Total Customers Complains
- Calculate Customers Complains Percentage

Y Accepted Campaigns Details :

- Total Campaign 1 Offers Acc
- Total Campaign 2 Offers Acc
- Total Campaign 3 Offers Acc
- Total Campaign 4 Offers Acc
- Last Campaign (Response)

Y Extract Products Details :

- Total Amount Spend On Fruits
- Total Amount Spend On Meat
- Total Amount Spend On Sweets
- Total Amount Spend On Gold

Y Calculate Some KPIs as :

- Total Customers
- Total Income
- Average each Customer Income

E-Commerce Customers Personality

SQL Extracted Insights Data Visualization & Storytelling

Customers Personality

Deal Purchases

5,208
Purchase

15.6% of Total Purchases

Store Purchases

12,970
Purchase

39.0% of Total Purchases

Website Purchases

9,150
Purchase

27.5% of Total Purchases

Catalog Purchases

5,963
Purchase

17.9% of Total Purchases

Total Income

116 M

Total Customers

2,240

TOP 5 Customer

	ID	Income \$
1	9432	666,666
2	1503	162,397
3	1501	160,803
4	5336	157,733
5	8475	157,243

Customer Complains

21 Complain | 0.9% of them Complain

Number of Accepted Campaigns Offer

This number represents the number of accepted individuals or the number of customers who accepted the offer for the advertising campaign, so we can see the extent of the impact of each campaign and make a comparison between them



Products Comparison

what are the most products in terms of the amount spent on them

Meat	75.8 %
Fruits	11.9 %
Sweets	12.3 %

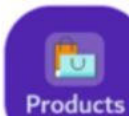


Customers Purchases Recency

Num of customers whose last purchase was during the previous month : 724



Amount Spend on Products



Products

374,310 \$ Meat	60,621 \$ Sweets	58,914 \$ Fruits
--------------------	---------------------	---------------------

Cookie Store Analysis

Exploartory Data Analysis using SQL to Extract Answers For Some Questions

REQUIRED QUESTIONS :



Calculate Each Customer Total Orders
(Showing His Name).



Calculate Each Customer Total Revenue
(Showing His ID, Name, City).



We Want to Know Total Quantity Of Each
Cookie (Showing Cookie ID, Cookie Name).



Calculate Total Quantity Of Each Cookie
(Cookie Name, Revenue, Cost, Profit).



What's Total Orders Number ?



What's Cookie Total Revenue ?



What's Cookie Total Profit ?



What's Cookie Total Cost ?



What's Total Orders Revenue ?

Cookie Store Analysis

SQL Extracted Qusetions Answers Data Visualization & Storytelling

SQL

Extracted Results Visualization
Cookies Store Analysis



- All Money amounts are in Dollar.
- Green is High, Yellow is Average and Red is Low
Tarek Ahmed Attia - Marketing Data Analyst



City Details :
Orders Revenue for each City

Seattle	8,418
Huntington	10,462
Green Bay	14,706
Salt Lake	14,870
Mobile	17,435

Customers Details :
Each Customer Total Orders

Tres Delicious	38
ABC Groceries	46
ACME Bites	49
Wholesome Foods	64
Park / Shop Stores	77

Products Details :
Each Product Quantity Sold, Revenue, Cost and Profit

Sugar	3,110 # Sold	3468 Revenue	1,734 Cost	1,734 Profit
Fortune Cookie	3,468 # Sold	9,330 Revenue	3,887 Cost	5,442 Profit
Snickerdoodle	2,751 # Sold	11,004 Revenue	4,126 Cost	6,877 Profit
White Chocolate Mac Nut	2,494 # Sold	12,470 Revenue	5,486 Cost	6,983 Profit
Oatmeal Raisin	2,489 # Sold	14,934 Revenue	6,844 Cost	8,089 Profit
Chocolate Chip	2,973 # Sold	14,685 Revenue	5,874 Cost	8,811 Profit

Facebook Campaigns Dashboard

Marketing Analytics Project – Data Visualization & Storytelling



Objective of Analysis :

I Want to Make a Visualization & Storytelling For 7 Facebook Campaigns Results on a Dashboard for ease to read the Campaigns Results

Steps :

- Data Ingestion
- Data Analyzing
- Data Visualization & Stortelling

Facebook Campaigns Dashboard

Marketing Analytics Project – Data Visualization & Storytelling

 Facebook
Campaign Analytics

 Campaign Info

Campaign - Ex 2

ID : 2384-3647-1568

 Add To Cart

30

Number

\$ 3,898

Conversion Value

 Purchases

35

Number

\$ 2,702

Conversion Value

Bid Strategy



Cost Cap

 Funnel

91

Lan Page Views

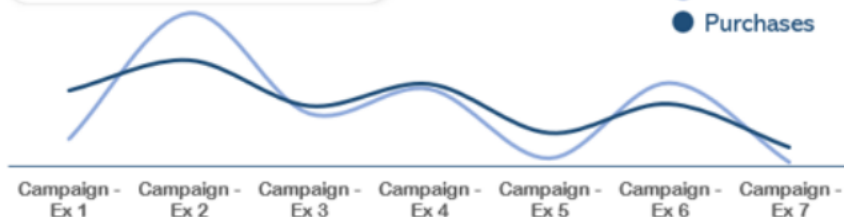
30

Add to Cart

35

Purchases

Add to Cart & Purchases



\$ 854
Budget

87,452
Impression

5,191
Reach

91
Land Views

3.16
ROAS

Made by /

Tarek Ahmed Attia
Junior Data Analyst

Campaigns Purchases



Twitter Campaigns Dashboard

Marketing Analytics Project – Data Visualization & Storytelling



Objective of Analysis :

I Want to Make a Visualization & Storytelling For 4 Twitter Campaigns Results on an Interactive Dashboard for ease to read the Campaigns Results

Steps :

- Data Ingestion
- Data Analyzing
- Data Visualization & Storytelling

Twitter Campaigns Dashboard

Marketing Analytics Project - Data Visualization & Storytelling



Web Scraping Project

Scrape Specific Data From Wuzzuf using Python – BeautifulSoup

16 lines (16 sloc) | 3.76 KB

Raw

Blame



Q Search this file...

	Job Title	Company	Location	Skills
1	Data Analyst	Gomla Market	Ameria, Alexandria, Egypt	Entry Level · Business Development · Logistics/Supply Chain · Analyst/Research · Analysis ·
3	Senior Data Analyst - Cairo	Confidential	Cairo, Egypt	Experienced · IT/Software Development · Analyst/Research · Engineering · Telecom/Techn
4	Senior Data Analyst	Alarabia Group	10th of Ramadan City, Cairo, Egypt	Experienced · IT/Software Development · Analyst/Research · Engineering · Telecom/Techn
5	Data Analyst	Hands of Hope Physical Therapy & Wellness	Maadi, Cairo, Egypt	Experienced · IT/Software Development · Analyst/Research · Analysis · Analyst · Data · Dat
6	Data Analyst	Gila Electric	New Cairo, Cairo, Egypt	Experienced · Accounting/Finance · IT/Software Development · Analyst/Research · Analysis
7	Data Analyst	Lecico Egypt	Alexandria, Egypt	Experienced · Analyst/Research · Analysis · Data Analyst · Data Analysis · Analyst · English
8	Data Analyst - Quality Assurance Department	IdealRatings	Heliopolis, Cairo, Egypt	Entry Level · Accounting/Finance · Quality · Analyst/Research · Finance · Quality Assurance
9	Senior Business Intelligence Analyst	Future Group	Dokki, Giza, Egypt	Experienced · IT/Software Development · Analyst/Research · Engineering · Telecom/Techn
10	Business System Analyst	The Food Lab	Maadi, Cairo, Egypt	Entry Level · Analyst/Research · Analysis · Computer Science · Data Analysis · Data Analyst
11	Senior Data Analyst	Z2 Data	Nasr City, Cairo, Egypt	Experienced · Analyst/Research · Data Analysis · Analysis · Administration · Computer Scie
12	Senior Financial Analyst	Future Group	Dokki, Giza, Egypt	Experienced · Accounting/Finance · Analyst/Research · Financial Data Analysis · Finance · a
13	Chargeback Analyst	WebBeds	Nasr City, Cairo, Egypt	Experienced · Accounting/Finance · Analyst/Research · Revenue Management · Financial A
14	Senior Treasury Analyst	TAQA Petroleum	Garden City, Cairo, Egypt	Experienced · Accounting/Finance · Analyst/Research · Finance · Financial Data Analysis · A
15	Data Analyst - Banha	Z2 Data	Banha, Qalubia, Egypt	Entry Level · Business Development · Analyst/Research · Analysis · Commerce · Computer
16	Senior Financial Data Analyst	Z2 Data	Nasr City, Cairo, Egypt	Experienced · Accounting/Finance · Analyst/Research · Analysis · Accounting · Financial · C

Freelancing Marketing Analytics Project

Facebook Page & Instagram Profile Two Quarters Comparison



Requirments :

A Client asked me to collect Data from Facebook Ads Manger to analyze Results of FB Page and Instagram Profile during two quarters of this year, in order to see the increase or decrease of the results and to make a Professional Report of the Result

Steps :

- Data Collecting from FB Ads Manger
- Data Visualization & Storytelling

Freelancing Marketing Analytics Project

Facebook Page & Instagram Profile Two Quarters Comparison

BEFORE AFTER COMPARISON PRODUCTS



Bike Sales Company Analysis

SQL Extract Results & Insights

Num Of Purchases For Each Occupation				
Management	Manual	Professional	Skilled Manual	Clerical
73	59	150	118	95

Relationship Between Miles & Num Of Purchases	
Miles	Num Of Purchases
0-1 Miles	207
1-2 Miles	83
2-5 Miles	95
5-10 Miles	77
10+ Miles	33

Num Of Purchases Per Region	
Region	Purchases
Europe	156
North America	220
Pacific	119

Num Of Purchases Per Gender	
Male	Female
243	252

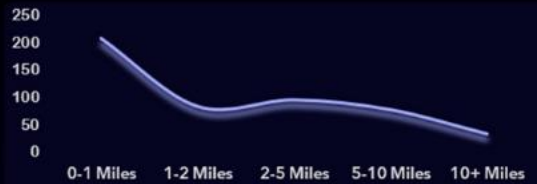
Bike Sales Company Analysis

SQL Extracted Questions Answers - Data Visualization & Storytelling

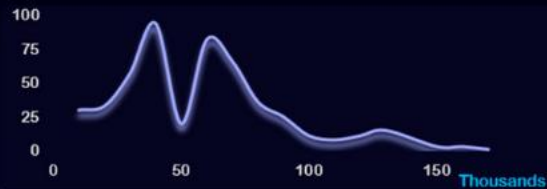


Bikes Sales

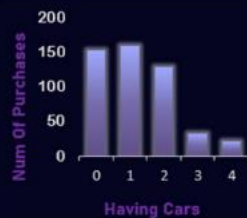
Relationship between Miles And Purchases



Relationship between Income And Purchases



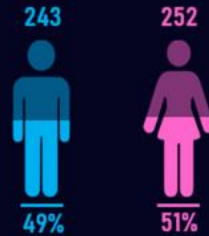
Car & Purchases



Home & Purchases



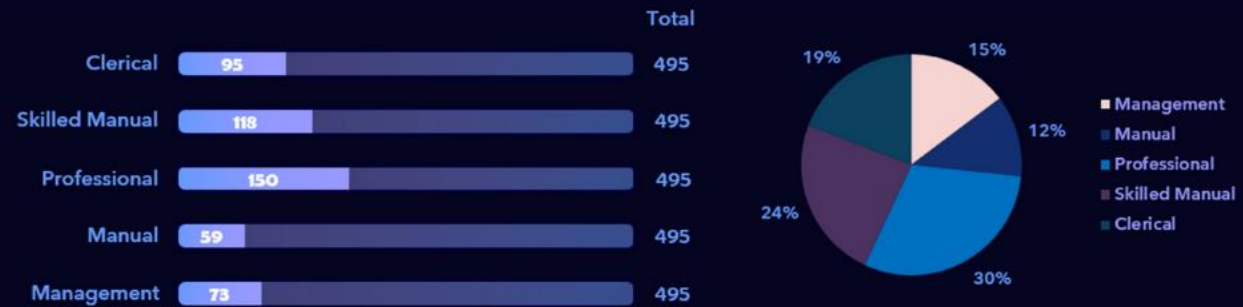
Purchases Per Gender



Purchases Per Region



Purchases Per Each Occupation



Performance Analysis Dashboard

Interactive Sales Dashboard For A Store



Objective of Analysis :

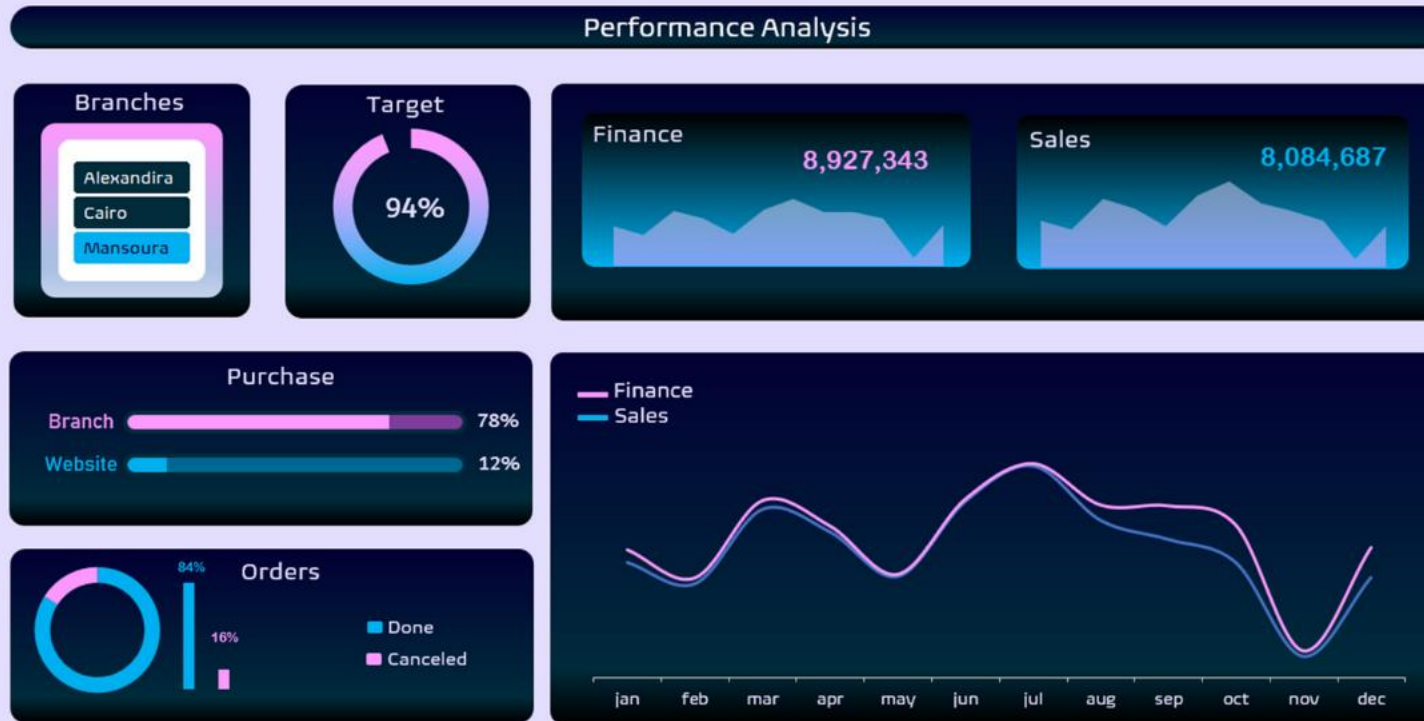
I Compared 3 Branches of a Store in terms of (Sales and Total Amount Paid, Target, Orders Details and Orders Status) to know the efficiency of each Branch

Steps :

- Data Ingestion
- Data Analyzing
- Data Exploration
- Data Visualization

Performance Analysis Dashboard

Interactive Sales Dashboard For A Store



Cairo

El Azhar Street

12 : 10

01014186474

Mansoura

El Emam Mohamed Abdo Street

10 : 8

01014186474

Alexandria

Saad Zagloul Street

1 : 11

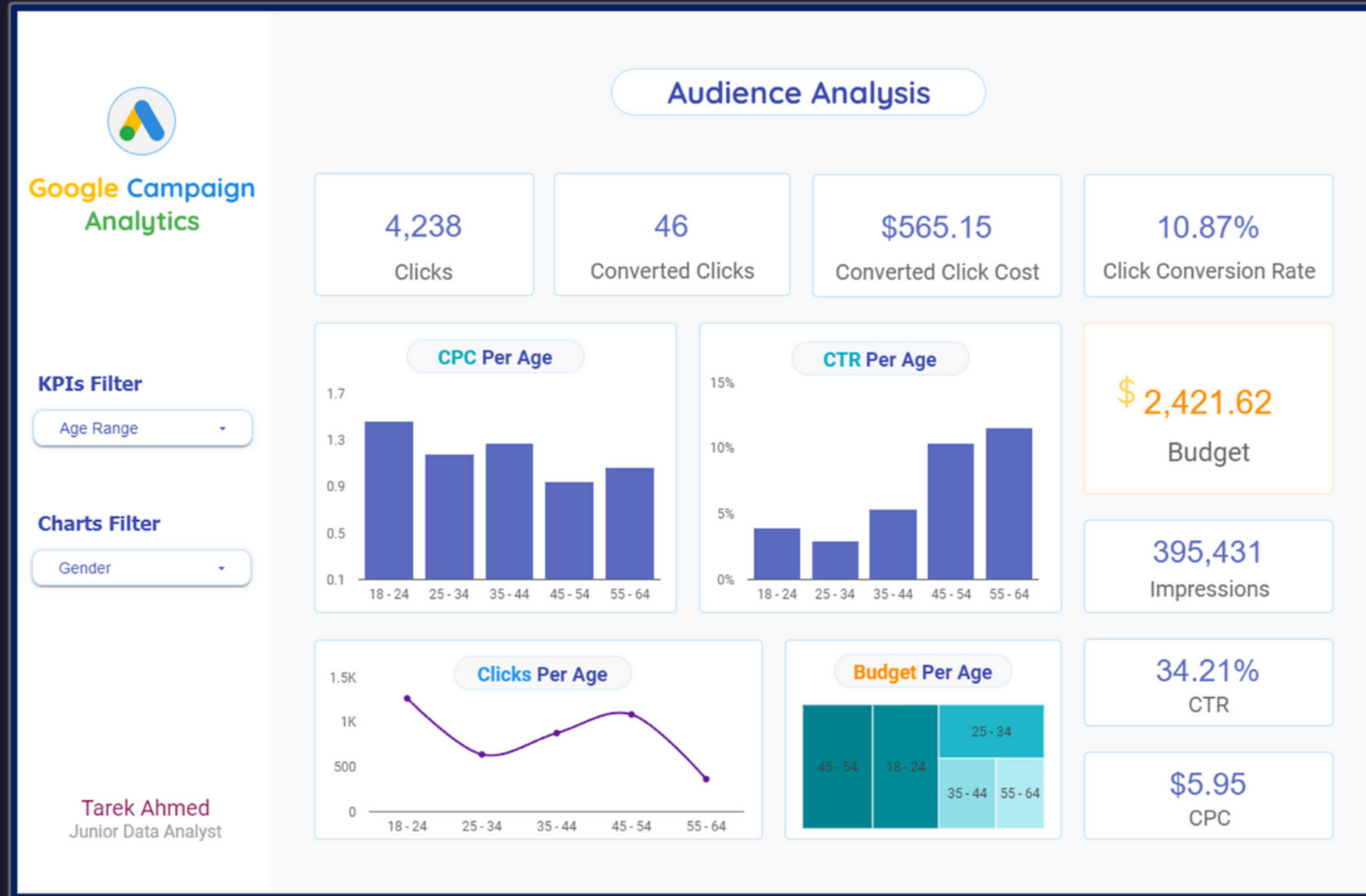
01014186474



www.Store.com

Campaign Audience Analytics

Marketing Analytics Project using Google Data Studio Report



E-Commerce Dashboard

Data Visualization using Power BI Software



Objective of Analysis :

In this Project, I Analyzed and Visualized the data of E-Commerce that have 4 Departments for 3 Years to make a Comparison between Departments.

Steps :

- Importing Data From MS Excel
- Data Visualization
- Publish The Dashboard Into Power BI Services

E-Commerce Dashboard

Data Visualization using Power BI Software



Covid-19 Analysis

Data Visualization using Power BI Software



Objective Of Analysis

In this Project, I Analyzed Corona Virus Data for Arab Countries to answer some questions and make a Comparison between them.

Steps :

- Importing Data From MS Excel
- Data Visualization
- Publish The Dashboard Into Power BI Services

Data Visualization using Power BI Software

Data Visualization using Power BI Software

515.6K
Total Cases

Total Cases

3.7M
Total Tests

Total Tests

442.2K
Total Recovered

Total Recovered

24.6K
Total Death

Total Death



Population

Area [Km²]

102M

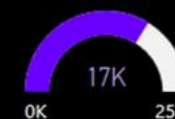


1M

Death

Hospitals

Homes



32.50%

67.50%

Recovered

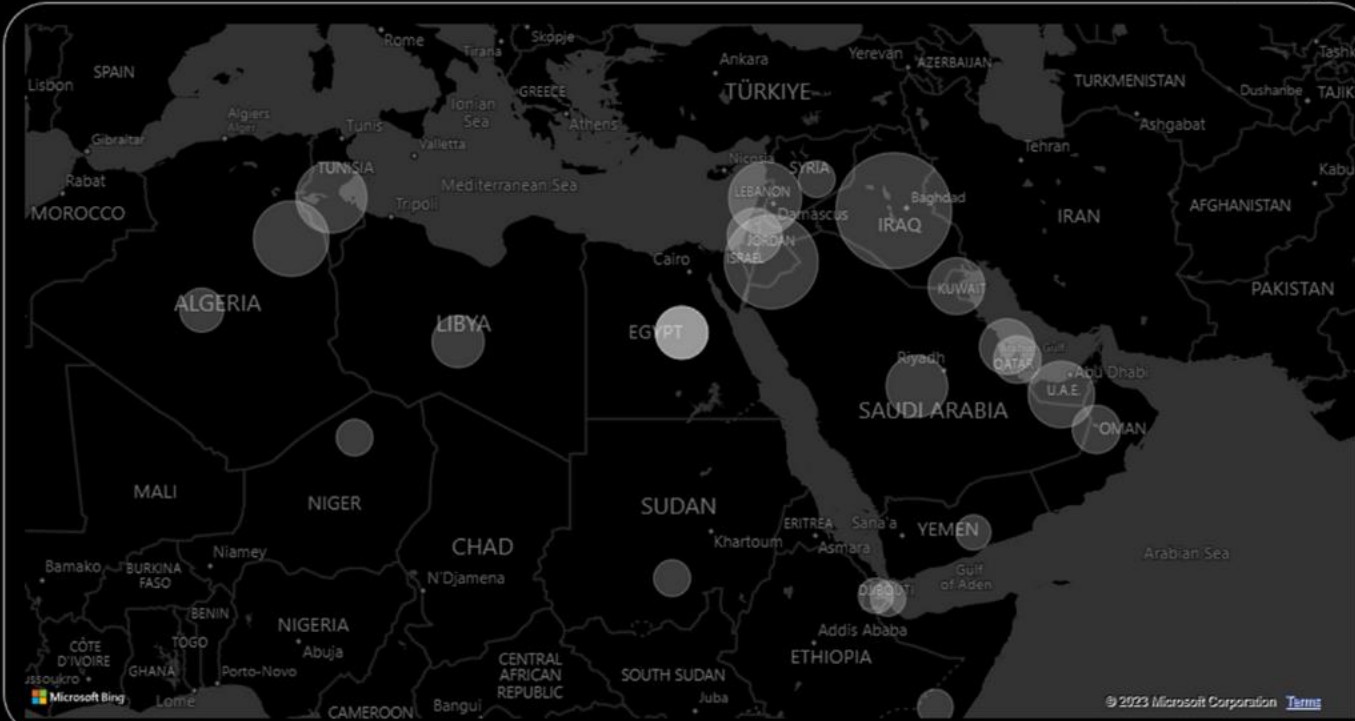
Hospitals

Homes



68.36%

31.66%



LapTop Store Dashboard

Interactive Sales Dashboard For A Store



Objective Of Analysis

I Analyzed the data of LapTop Store and create an Interactive Dashboard to make a Comparison between 7 Types of LapTop to know what type is the most selling

Steps :

- Data Ingestion
- Data Analyzing
- Data Exploration
- Data Visualization

LapTop Store Dashboard

Interactive Sales Dashboard For A Store



USA Store Analysis

Python Data Analysis Project



Objective Of Analysis

In this Project, I used Python to Perform Exploratory Data Analysis (EDA) to USA Store to answer some question and do Data Visualization to the Information.

Steps :

- Import Data from Excel
- Data Cleaning
- Data Exploration
- Data Visualization

USA Store Analysis

Python Data Analysis Project

- More categories details (Each sub category sales in each category)

```
In [33]: fur_subcat = ["Bookcases", "Chairs", "Furnishings", "Tables"]
fur_subcat_sales = [114880.00, 328449.10, 91705.16, 206965.53]
office_subcat = ["Appliances", "Art", "Binders", "Envelopes", "Fasteners", "Labels", "Paper", "Storage", "Supplies"]
office_subcat_sales = [107532.16, 27118.79, 203412.73, 16476.40, 3024.28, 12486.31, 78479.21, 223843.61, 46673.54]
tech_subcat = ["Accessories", "Copiers", "Machines", "Phones"]
tech_subcat_sales = [167380.32, 149528.03, 189238.63, 330007.05]
colors = ["#C85250", "#E5C101", "#E5C101", "#18A558"]
c1 = ["#E5C101", "#18A558", "#C85250", "#E5C101"]
c2 = ["#E5C101", "#C85250", "#18A558", "#C85250", "#C85250", "#C85250", "#E5C101", "#18A558", "#E5C101"]
c3 = ["#E5C101", "#E5C101", "#E5C101", "#18A558"]
```

```
In [34]: plt.style.use('seaborn')
fig, ax = plt.subplots(nrows = 3, ncols = 1)
ax[0].bar(fur_subcat, fur_subcat_sales, width = .1, color = c1)
ax[0].set_title("Furniture")
ax[1].bar(office_subcat, office_subcat_sales, width = .27, color = c2)
ax[1].set_title("Office Supplies")
ax[2].bar(tech_subcat, tech_subcat_sales, width = .1, color = c3)
ax[2].set_title("Technology")
fig.tight_layout(pad = 1)
plt.show()
```



Flight Company Analysis

Python Data Analysis Project



Objective Of Analysis

In this Project, I used Python to Analyze Flight Company Data to answer some question then Visualize the Results.

Steps :

- Import Data from Excel
- Data Cleaning
- Data Exploration
- Data Visualization

Flight Company Analysis

Python Data Analysis Project

Q / Which Agency Book The Most Flights ?

The Most Flights Booking By | Rainbow Agency

```
In [48]: data["agency"].unique()
```

```
Out[48]: array(['FlyingDrops', 'CloudFy', 'Rainbow'], dtype=object)
```

```
In [50]: data["agency"].value_counts().sort(ascending = False)
```

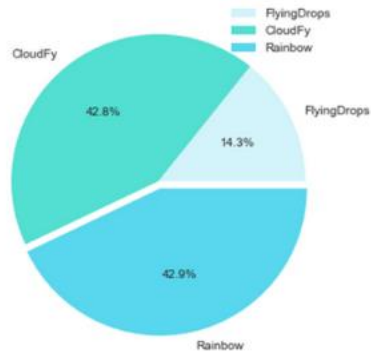
```
Out[50]: Rainbow      116752  
CloudFy      116378  
FlyingDrops    38758  
Name: agency, dtype: int64
```

```
In [51]: agencies = ['FlyingDrops', 'CloudFy', 'Rainbow']  
num_of_booking_flights = [38758, 116378, 116752]
```

```
In [54]: np.max(num_of_booking_flights)
```

```
Out[54]: 116752
```

```
In [64]: plt.style.use('seaborn')  
colors = ["#D3F4FB", "#53DFD1", "#59D7EE"]  
explode = [0, 0, 0.05]  
plt.pie(num_of_booking_flights, data = data, colors = colors, labels = agencies, explode = explode, autopct = "%1.1f%%")  
plt.legend()  
plt.show()
```



Corona Virus Analysis

Python Data Analysis Project



Objective Of Analysis

In this Project, I used Python to Analyze Corona Virus Data for Arab Countries to explain more Details.

Steps :

- Import Data from Excel
- Data Cleaning
- Data Exploration
- Data Visualization

Corona Virus Analysis

Python Data Analysis Project

```
In [16]: data[data["Total Recovered"] == data["Total Recovered"].max()]
```

```
Out[16]:
```

	Country	Area	Population	Total Cases	Total Tests	Total Death	Total Recovered	Rec Home	Rec Hos	Death At Homes	Death At Hospitals
8	Iraq	435244	402222493	2329313	18631073	25221	2302855	802855	1500000	10000	15221

```
In [17]: data[data["Total Recovered"] == data["Total Recovered"].min()]
```

```
Out[17]:
```

	Country	Area	Population	Total Cases	Total Tests	Total Death	Total Recovered	Rec Home	Rec Hos	Death At Homes	Death At Hospitals
21	Union Of Comoros	2235	869601	8100	113212	160	7933	3933	4000	90	70

```
In [18]: round(data["Total Recovered"].mean(), 2)
```

```
Out[18]: 555742.68
```

Calc Percentage Of Death & Recovered At (Hos & Home)

```
In [19]: death = [death for death in data["Total Death"]]
death_at_home = [death_at_home for death_at_home in data["Death At Homes"]]
death_at_hos = [death_at_hos for death_at_hos in data["Death At Hospitals"]]
```

```
In [20]: death_hos_per = [ round((n1/n2)*100) for n1, n2 in zip(death_at_hos, death)]
death_home_per = [ round((n1/n2)*100) for n1, n2 in zip(death_at_home, death)]
```

```
In [21]: rec = [rec for rec in data["Total Recovered"]]
rec_at_home = [rec_at_home for rec_at_home in data["Rec Home"]]
rec_at_hos = [rec_at_hos for rec_at_hos in data["Rec Hos"]]
```

```
In [22]: rec_hos_per = [ round((n1/n2)*100) for n1, n2 in zip(rec_at_hos, rec)]
rec_home_per = [ round((n1/n2)*100) for n1, n2 in zip(rec_at_home, rec)]
```

```
In [23]: data.insert(11, column = "Per of Rec At Hos", value = rec_hos_per)
data.insert(12, column = "Per of Rec At Home", value = rec_home_per)
data.insert(13, column = "Per of Death At Hos", value = death_hos_per)
data.insert(14, column = "Per of Death At Home", value = death_home_per)
```

REQUIRED QUESTIONS :



Create a Revenue Column to Calculate each Order Revenue In Orders Table



Calculate Total Profit In 2020



Create a Cost Column to Calculate each Order Cost In Orders Table



Calculate Total Orders & Total Customers In 2020



Create a Profit Column to Calculate each Order Profit In Orders Table



Calculate Total Units Sold In 2020



What is Total Number of Customers ?



Calculate Total Number Of Orders ?



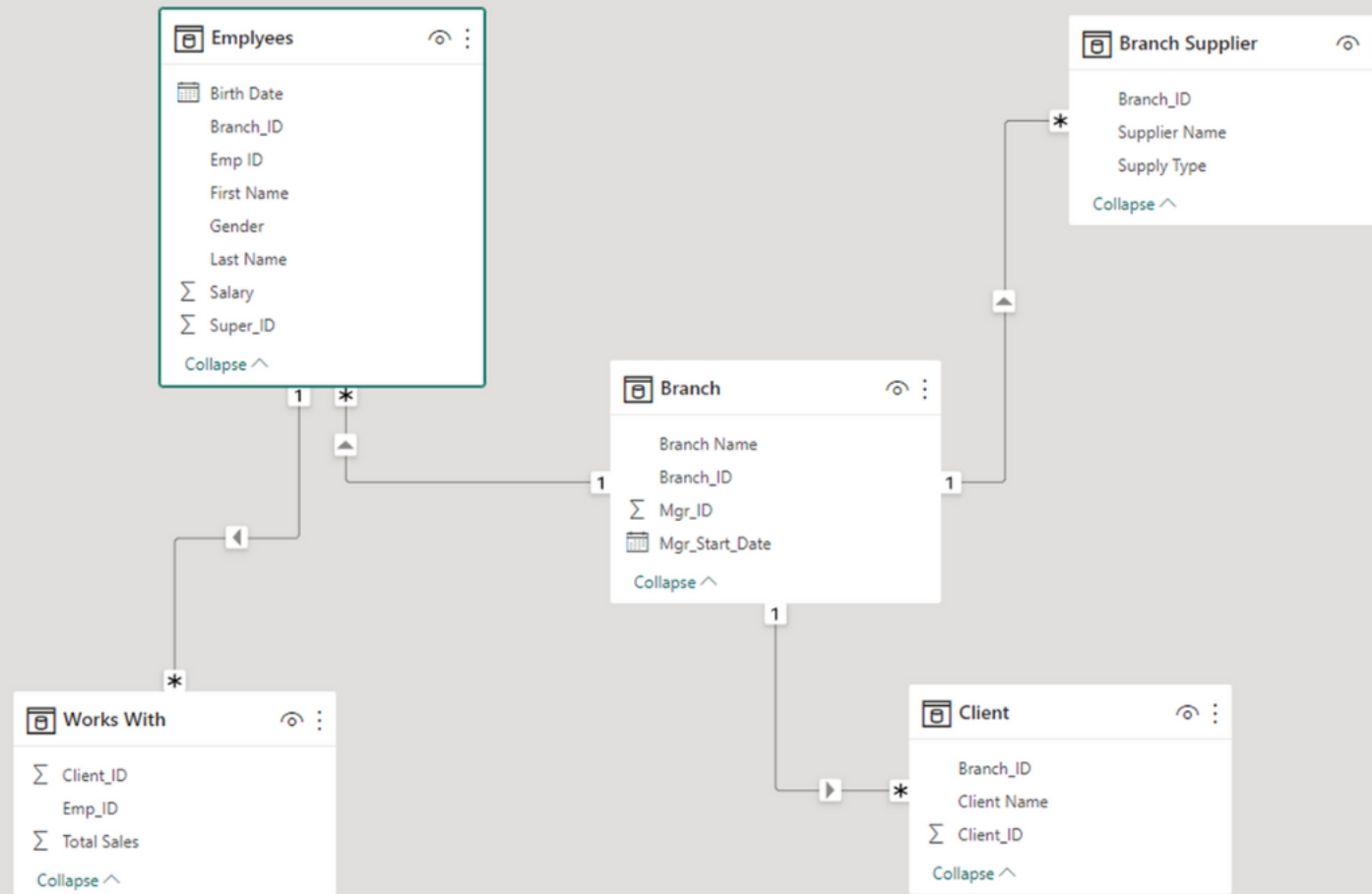
Create a Separated Tabel for the Year 2020.



Calculate Total Revenue In 2020

Data Modeling

Tables Relationships using Power BI Software



Airlines Company Analysis

Pivot Table Analysis Results Data Visualization



Objective Of Analysis

In this Project, I used Python to Perform Exploratory Data Analysis (EDA) to USA Store to answer some question and do Data Visualization to the Information.

Steps :

- Data Analyzing
- Data Exploration
- Data Visualization

Airlines Company Analysis

Pivot Table Analysis Results Data Visualization



Airlines

Rate



Online Boarding



Baggage Handling



Checkin Service

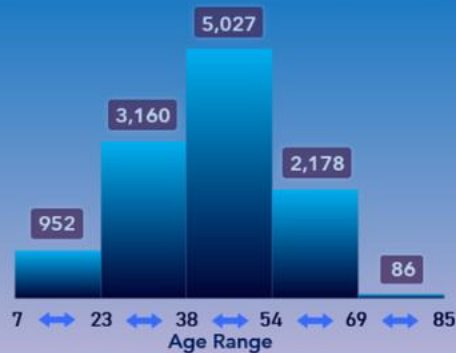


Gate Location



Departure/Arrival
Time Convenient

Satisfied Customers | Age



Satisfied Customer | Type Of Travel

Personal



Satisfied
793
10%
Of Total
Customers

Business



Satisfied
10,610
59%
Of Total
Customers

Satisfied Customer | Type Of Class

Business

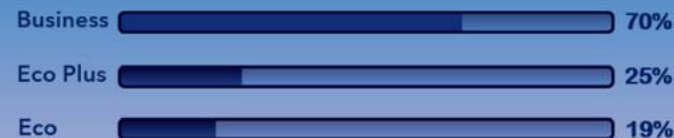
Satisfied **8,686**

Eco Plus

Satisfied **475**

Eco

Satisfied **2,242**



Freelancing Project

Depression Causes Analysis Report



Objective Of Analysis

A Client Asked me to Analyze the Causes of Depressions among youth and make a Report to this Results.

Steps :

- Data Analyzing
- Data Cleaning
- Data Exploration
- Data Visualization

Freelancing Project

Depression Causes Analysis Report

Causes of Depressions among Youth

Female

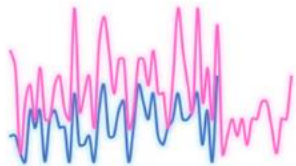
Male

Poor Appetite

Above or Equal Rate 5
Total Number : 37

Under Rate 5
Total Number : 21

Relation Between M & F

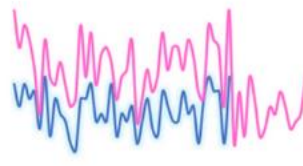


Feeling Bad Yourself

Above or Equal Rate 5
Total Number : 27

Under Rate 5
Total Number : 31

Relation Between M & F



Concentrating on Things

Above or Equal Rate 5
Total Number : 49

Under Rate 5
Total Number : 9

Relation Between M & F

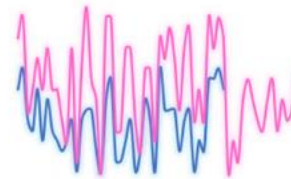


Moving / Speaking Slowly

Above or Equal Rate 5
Total Number : 35

Under Rate 5
Total Number : 23

Relation Between M & F

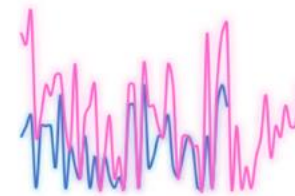


Hurting Yourself

Above or Equal Rate 5
Total Number : 26

Under Rate 5
Total Number : 32

Relation Between M & F



Analytical Presentation

Competition Teams Analysis and Presentation





FEEL FREE TO CONTACT ME:

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- ✓ **phone:** 01023360497
- ✓ **whatsapp:** 01120055679
- ✓ **linkedin:** mohamed-fouad-08a245266